

# AI Phase Plan for Convergent Regulatory Changes in Spontaneous Decidualization

## Executive summary

The best first implementation phase is **not** a fully general three-species foundation-model project. The public data landscape is asymmetric: humans have rich cycle-resolved transcriptomic and chromatin data, mice have strong decidualization atlases and are the best immediate non-spontaneous-decidu-alization outgroup, bats now have pregnancy-stage maternal-fetal interface data, and the Egyptian spiny mouse still appears to have much thinner public omics coverage centered on genome and transcriptome references rather than uterus-specific single-cell or chromatin assays. Because spontaneous decidualization is reported in humans, some bats, and the spiny mouse, while ordinary laboratory mice decidualize only after embryo-driven signaling, the most actionable phase is a **human-first, mouse-anchored, bat-validation, spiny-mouse-sequence-first** design. <sup>1</sup>

For implementation, make **scRNA-seq / snRNA-seq the backbone**, use **bulk RNA-seq** to densify time axes and perturbation signals, add **ATAC-seq and receptor cistromes** as causal priors where available, and treat **spatial data** mainly as a validation layer. On the AI side, the highest-value stack is: **scGPT or scvi-tools/scANVI for harmonization and label transfer**, **Geneformer for human-centric regulator prioritization**, **GENIE3 as the expression-only GRN baseline**, **LINGER only where paired RNA+ATAC exists or pseudo-multiome is credible**, and **Enformer/Borzoï as the reproducible open sequence baselines**, with **AlphaGenome** as an optional frontier model if API access and license constraints fit the project. <sup>2</sup>

Most public source data for this phase are housed in NCBI <sup>3</sup> archives such as GEO and SRA, or in EMBL-EBI <sup>4</sup> resources such as ArrayExpress and BioStudies. <sup>5</sup>

## Dataset registry

The registry below is optimized for an AI narrowing phase rather than for a full evolutionary atlas. The key design choice is to separate **discovery datasets** from **validation datasets**. In practice, human and mouse will carry most of the discovery burden; bat data will mostly validate whether shortlisted stromal and maternal-fetal interface regulators recur in an independently evolved lineage; spiny mouse public resources are currently most useful for genome-aware ortholog and regulatory-element mapping, unless you locate additional uterus-specific data outside the sources reviewed here. <sup>6</sup>

| Species | Accession / resource | Assay                                                 | Sample count and stage info                                                                                                             | Recommended use in this phase                                                                                                    | Primary source |
|---------|----------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------|
| Human   | <b>GSE111976</b>     | scRNA-seq, in vivo endometrium                        | 19 biopsies spanning menstrual-cycle days 4–27                                                                                          | Core reference for epithelial, stromal, endothelial, and immune cell-state harmonization across the natural cycle                | 7              |
| Human   | <b>GSE127918</b>     | scRNA-seq, biopsies + in vitro stromal time course    | 17 libraries total: 6 mid-luteal biopsy samples plus decidualization time-course libraries including D0, D2, D4, D6, D8, and withdrawal | Primary human decidual-trajectory dataset for stromal state transitions and branching into decidual versus senescent programs    | 8              |
| Human   | <b>GSE183771</b>     | scATAC-seq, in vivo endometrium                       | 12 timed biopsies, 13 libraries, spanning proliferative and secretory stages labeled P7 to S25                                          | Primary human chromatin dataset for motif activity, TF prioritization, and linking accessibility changes to stromal state shifts | 9              |
| Human   | <b>GSE205477</b>     | bulk RNA-seq, primary human endometrial stromal cells | 12 libraries from 3 donors; pre-decidual versus decidual, with and without TRIM28 knockdown                                             | Perturbation-informed expression prior for candidate regulator ranking and perturbation-prediction benchmarking                  | 10             |

| Species | Accession / resource          | Assay                                               | Sample count and stage info                                                                                     | Recommended use in this phase                                                                               | Primary source |
|---------|-------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------|
| Human   | <b>GSE205473</b>              | ATAC-seq, primary human endometrial stromal cells   | 8 libraries; pre-decidual versus decidual, with and without TRIM28 knockdown                                    | Perturbation-informed accessibility prior for validating TF-RE-TG hypotheses                                | 11             |
| Human   | <b>GSE62475 / PRJNA264199</b> | PGR-A / PGR-B ChIP-seq in human stromal cells       | Indexed page confirms decidualization-associated cistrome; sample count not explicit in the page excerpt        | Receptor-cistrome validation for progesterone-response edges in shortlisted human GRNs                      | 12             |
| Human   | <b>E-MTAB-9260</b>            | Visium spatial transcriptomics, in vivo endometrium | Visium in vivo dataset; indexed sources describe it as a spatial endometrial reference used in later atlas work | Spatial validation of predicted stromal and epithelial states and ligand-receptor neighborhoods             | 13             |
| Human   | <b>GSE180485</b>              | bulk RNA-seq, secretory/luteal endometrium          | 36 endometrial biopsies from the secretory phase                                                                | Timing model and pseudotime calibration for luteal-stage ordering and pseudobulk trajectory smoothing       | 14             |
| Human   | <b>GSE168405</b>              | scRNA-seq, endometrial assembloids                  | 9 libraries across day 0, day 4, and day 4 plus dasatinib                                                       | Controlled perturbation-like system for testing senescence-linked state abstraction and cell-state transfer | 15             |

| Species     | Accession / resource   | Assay                                       | Sample count and stage info                                                                                                     | Recommended use in this phase                                                                   | Primary source |
|-------------|------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------|
| Bat         | <b>GCF_021234435.1</b> | Jamaican fruit bat genome assembly          | Reference assembly                                                                                                              | Required for sequence extraction, synteny, motif scanning, and candidate-element ranking in bat | 16             |
| Bat         | <b>PRJNA1251670</b>    | snRNA-seq, maternal-fetal interface         | Pregnancy-state bat placenta / maternal-fetal interface data; exact run count should be enumerated from the BioProject manifest | Primary bat single-nucleus validation dataset for decidua/stromal immune programs               | 17             |
| Bat         | <b>PRJNA1251235</b>    | single-cell dataset listed by the bat paper | Pregnancy-state; exact run count should be enumerated from the BioProject manifest                                              | Optional bat cell-state validation dataset if organoid or dissociated-cell profiles are needed  | 18             |
| Bat         | <b>PRJNA1251203</b>    | bulk RNA-seq listed by the bat paper        | Pregnancy-state; exact run count should be enumerated from the BioProject manifest                                              | Pseudobulk validation and differential-expression bridge when cell-state resolution is noisy    | 18             |
| Spiny mouse | <b>GCA_029890205.1</b> | chromosome-scale genome assembly            | One reference assembly, ~2.3 Gb                                                                                                 | Primary spiny-mouse sequence reference for ortholog mapping and regulatory-sequence scoring     | 19             |

| Species        | Accession / resource           | Assay                                                 | Sample count and stage info                                                                                                                                             | Recommended use in this phase                                                                                                                                     | Primary source |
|----------------|--------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Spiny mouse    | <b>PRJNA182705</b>             | de novo transcriptome assembly                        | Transcriptome built from 15 organ types; cycle stage not stated                                                                                                         | Annotation support and transcript evidence for ortholog and motif analyses                                                                                        | 20             |
| Spiny mouse    | <b>Gap in reviewed sources</b> | uterus-specific scRNA / ATAC / ChIP / spatial         | In the indexed sources reviewed, I did not find a clearly public uterus-specific spiny-mouse single-cell or chromatin accession                                         | Treat spiny mouse as sequence-first in Phase 1; add wet-lab or new data-collection workstream if this species must contribute transcriptomic evidence immediately | 21             |
| Mouse fallback | <b>E-MTAB-11491</b>            | scRNA-seq across estrus cycle and decidualization     | Stage-resolved multi-organ study including all four estrus stages and pregnant day 5.5 decidualized uterus; enumerate exact uterus sample count from accession manifest | Best immediate non-spontaneous-decidu-alization reference for cell-type harmonization and evolutionary contrast                                                   | 22             |
| Mouse fallback | <b>GSE226417</b>               | scRNA-seq, early pregnancy decida / uterus / placenta | 11 samples spanning E5.5–E10.5 plus control                                                                                                                             | High-value decidual stromal state atlas for identifying transition-associated cell states and immune-featured stromal cells                                       | 23             |

| Species        | Accession / resource | Assay                                              | Sample count and stage info                                                                               | Recommended use in this phase                                                                                       | Primary source |
|----------------|----------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------|
| Mouse fallback | <b>GSE226429</b>     | bulk RNA-seq, in vitro decidualization time series | 6 samples from control through day 5                                                                      | Clean time-course data for dynamic GRN inference and cross-checking candidate temporal regulators                   | 24             |
| Mouse fallback | <b>GSE122376</b>     | bulk RNA-seq across three decidualization models   | Three replicates per group; natural pregnancy day 8, artificial decidualization day 8, and in vitro day 4 | Robust mouse bulk reference for consensus decidual signatures across model systems                                  | 25             |
| Mouse fallback | <b>GSE205480</b>     | scRNA-seq, perturbation                            | 2 pooled genotype libraries at pregnancy day 3.5                                                          | Small but useful perturbation dataset for checking whether shortlisted regulators shift uterine stromal composition | 26             |
| Mouse fallback | <b>E-MTAB-12105</b>  | Visium spatial transcriptomics                     | Young diestrus and aged uterine / ovarian tissues; enumerate exact sample count from accession            | Spatial anchor for anatomy-aware validation and stromal neighborhood analyses                                       | 27             |

Two practical consequences follow from this registry. First, **human and mouse can support a serious discovery pipeline immediately**. Second, **bat and spiny mouse are asymmetric**: bat has usable pregnancy-state interface data but not a rich public cycle atlas in the reviewed sources, while spiny mouse currently contributes most strongly through its genome and transcript reference. That asymmetry is why the initial AI phase should optimize for **search-space narrowing**, not final proof of convergence. 28

## Model and library stack

The highest-confidence stack is to use a **two-layer AI design**. Layer one learns cross-species cell states and candidate regulators from expression and, where available, chromatin. Layer two scores orthologous or syntenic regulatory sequences for their predicted impact on accessibility and transcription. The key practical point is that data asymmetry argues for **fine-tuning and transfer**, not from-scratch pretraining. Human-only models such as Geneformer are especially useful for ranking human decidual regulators and then projecting that shortlist into bat and spiny ortholog space; models like scGPT and scvi-tools are better suited to integration and cross-dataset harmonization. <sup>29</sup>

| Model / tool               | Purpose in this project                                                                     | Typical input → output                                                                   | Intended use here                                                                                                       | Maturity       | Primary source |
|----------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------|----------------|
| <b>scGPT</b>               | Cell and gene representation learning; integration; annotation; perturbation-style modeling | count matrix / AnnData → embeddings, labels, predicted states                            | Use as a flexible foundation model for cross-dataset stromal-state harmonization after ortholog reduction               | Research-ready | <sup>30</sup>  |
| <b>Geneformer</b>          | Context-aware human gene-network representation and in silico perturbation                  | ranked expressed genes per cell → embeddings, perturbed embeddings, candidate regulators | Use first on human stromal and decidual cells to rank regulators, then project the shortlist to bat and spiny orthologs | Research-ready | <sup>31</sup>  |
| <b>scvi-tools / scANVI</b> | Probabilistic integration and label transfer                                                | AnnData counts → latent space, integrated embeddings, semi-supervised labels             | Recommended backbone for donor-aware integration, batch correction, and hierarchical cell-type transfer                 | Mature         | <sup>32</sup>  |

| Model / tool              | Purpose in this project                                                   | Typical input → output                                                          | Intended use here                                                                                                                          | Maturity                              | Primary source |
|---------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------|
| <b>LINGER</b>             | TF-RE-TG GRN inference from paired RNA+ATAC plus motifs                   | paired single-cell multiome → regulatory edges and subnetworks                  | Use only where truly paired multiome exists or where pseudo-multiome aggregation is credible; otherwise do not make it the sole GRN engine | Research-ready, specialized           | 33             |
| <b>GENIE3 / dynGENIE3</b> | Expression-only GRN inference; time-series variant available              | bulk or pseudobulk expression → TF-target importance scores                     | Make this the baseline GRN method, especially for human and mouse time-course or pseudobulk analyses                                       | Mature                                | 34             |
| <b>Enformer</b>           | Long-range sequence-to-expression prediction                              | ~200 kb DNA context → functional genomic tracks / variant effects               | Open, reproducible baseline for testing candidate enhancer and promoter sequence changes                                                   | Mature                                | 35             |
| <b>Borzoï</b>             | Sequence-to-RNA-seq and regulatory-track prediction                       | 524 kb DNA context → RNA-seq coverage and additional regulatory signals         | Strong open sequence model for promoter-enhancer effect prediction and element prioritization                                              | Research-ready / strong open baseline | 36             |
| <b>AlphaGenome</b>        | Unified multimodal sequence model with long context and base-level output | 1 Mb DNA → thousands of genomic tracks, splicing and variant-effect predictions | Optional frontier model for final rescoring of top candidate elements if API access is feasible                                            | Frontier; API preview                 | 37             |

A practical recommendation follows from the table: **do not center Phase 1 on LINGER or AlphaGenome**. LINGER is strongest when you truly have paired RNA+ATAC, which the current public focal-species set only



partially supports, and AlphaGenome is compelling but currently exposed through an API preview from Google DeepMind <sup>38</sup>, which can complicate reproducibility and non-commercial restrictions. Make **scvi-tools + scGPT/Geneformer + GENIE3 + Enformer/Borzoï** the default stack, and treat LINGER and AlphaGenome as optional accelerants. <sup>39</sup>

## Ortholog strategy

Use a **three-tier ortholog map**, not a single collapsed table. Tier one is the **strict one-to-one backbone**, which should drive cross-species integration, cell-state transfer, and the primary convergence tests. Tier two is a **permissive orthogroup layer** for genes that are one-to-many or many-to-many in bats or spiny mouse. Tier three is a **regulatory-syntenic layer** for elements, because element-level conservation often fails long before gene-level orthology becomes ambiguous. This separation matters because Ensembl explicitly distinguishes one-to-one, one-to-many, and many-to-many orthologues, and also provides orthology quality controls such as gene-order conservation and whole-genome-alignment scores. <sup>40</sup>

The concrete mapping rule should be: **human as anchor species**, then map to mouse and bat through Ensembl Compara/BioMart where possible, and use g:Orth as a batch identifier-conversion and cross-check layer because it is built from Ensembl ortholog mappings. For the spiny mouse, which is the least mature of the focal species in public annotation terms, use OrthoDB as the broad-coverage fallback and then validate critical mappings with reciprocal best hits against the latest assembly and transcriptome. For primary discovery, exclude unresolved paralogs from the strict backbone and analyze them only in a secondary family-aware pass. <sup>41</sup>

At the regulatory level, compare **orthologous genes directly** but compare **candidate regulatory elements by synteny, motif grammar, and TE family**, not just by exact sequence alignment. That is especially important here because both comparative uterine transcriptomics and human cyclical chromatin studies point to heavy transposable-element involvement in endometrial regulatory remodeling, making TE-aware convergence tests more informative than strict basewise conservation alone. <sup>42</sup>

## Implementation workflow

Start with a **processed-matrix-first ingestion strategy**. Pull matrices and metadata first from GEO, SRA-derived processed files, and ArrayExpress/BioStudies, and defer raw FASTQ remapping to datasets that are chromatin-based, explicitly perturbational, or missing reliable processed outputs. For 10x-style single-cell data, either use the provided processed matrices or reprocess with Cell Ranger; its current documentation states that Cell Ranger uses the STAR aligner under the hood. For bulk RNA-seq, use Salmon for fast transcript quantification unless you specifically need genome-coordinate alignments for splice-aware inspection or track generation. <sup>43</sup>

For QC, I would use these **starting parameters**. For scRNA/snRNA: keep cells with at least 500 genes detected in human and mouse, relax to 300 for low-depth bat or pilot spiny analyses, remove cells above 15% mitochondrial reads in human and mouse and above 20% only in exploratory non-model analyses, and remove doublets with an expected rate of 5–10%. Build pseudobulk profiles only from donor × stage × cell-type bins with at least 20 cells. Keep raw counts in `h5ad` and write one donor-harmonized metadata table as the system of record.

Then build the integration in two passes. Pass one is a **species-specific atlas pass** using Scanpy plus scvi-tools to create high-confidence cell labels for human and mouse separately. Pass two is a **cross-species reduced-gene pass** using the strict one-to-one ortholog backbone and a stromal-focused state ontology: stromal fibroblast, pre-decidual, decidual, senescent-decidual, endothelial/perivascular, immune, and glandular epithelial. Use scANVI or a similar semi-supervised model for label transfer, and deliberately hold out one species at a time to test whether the latent space generalizes. <sup>44</sup>

For model fitting, keep the phase narrow. Fine-tune scGPT on ortholog-reduced stromal and neighboring cell states across human, mouse, and bat. Run Geneformer primarily on the human stromal subset to produce a ranked list of candidate upstream regulators under decidual progression and senescence branching. Use ranked outputs rather than end-to-end classifier predictions as the handoff into downstream GRN and sequence analyses. Because Geneformer is human-pretrained, treat its cross-species value as **shortlist generation through ortholog projection**, not as a literal multi-species foundation model. <sup>45</sup>

For GRN inference, run **GENIE3 first** on stromal-cell pseudobulk and time-ordered human and mouse series, then intersect its edges with human ATAC and receptor-cistrome support. Use LINGER only when you can credibly construct matched RNA and chromatin representations for the same cell states; otherwise it should remain a secondary method. The output of this layer should be a TF-target table and, where possible, a TF-regulatory-element-target table for each species, plus a merged comparison table keyed by strict orthologs. <sup>46</sup>

For sequence-based regulatory prediction, take the top 200–1,000 candidate elements from the GRN layer and score them with Enformer or Borzoi using human, bat, spiny-mouse, and mouse sequences in matched syntenic windows. Use three perturbation modes: **native scoring**, **motif ablation**, and **sequence swap**. AlphaGenome should only be brought in at the end to rescore the top shortlist if preview API access is acceptable. This phase should explicitly test whether candidate elements near orthologous genes show concordant predicted effects in humans and bats, and then in spiny mouse where synteny is clear, while diverging in the mouse outgroup. <sup>47</sup>

For convergence testing, use a layered design. Test convergence at the **gene level** by overlap and rank aggregation of differential expression and regulator scores. Test convergence at the **network level** by overlap of regulons and TF-target edges. Test convergence at the **element level** by shared predicted sequence effects, shared motif turnover, and TE-family enrichment. Finally, add a **phylogenetic comparative layer** only after expanding the comparison set beyond the focal trio, because strong claims about convergence are much more credible with additional mammalian outgroups such as rabbit, opossum, and non-menstruating rodents. <sup>48</sup>

## Evaluation and deliverables

The evaluation should be built around five metrics families. For **cross-species transfer**, report macro-F1, balanced accuracy, and confusion matrices for hierarchical cell labels under leave-one-species-out transfer. For **trajectory fidelity**, compare pseudotime orderings to known cycle or experimental order using Spearman correlation and donor-aware concordance. For **GRN quality**, use AUPR or AUROC against edges supported by ATAC, ChIP/CUT&RUN, and literature-backed decidual regulators. For **sequence scoring**, report correlation between predicted and observed accessibility or expression shifts, plus AUROC for held-

out positive regulatory elements. For **convergence**, report empirical permutation p-values, effect sizes, and FDR-adjusted q-values for shared genes, regulons, motifs, and TE families. <sup>49</sup>

A strong validation plan should include at least three orthogonal checks. First, test whether candidate regulators predict the known effects in human and mouse perturbation datasets such as the TRIM28 series. Second, confirm that top predicted regulators are supported by motif accessibility or receptor cistrome evidence in human endometrium. Third, confirm that top sequence-level candidates remain prioritized when using both an open model baseline and, if available, AlphaGenome. If you can add one wet-lab follow-on after this computational phase, the best choices are CRISPRi reporter assays or targeted enhancer-reporter swaps in human stromal cells. <sup>50</sup>

The deliverables should be concrete and machine-usable. I would require: a `dataset_registry.csv`; an `ortholog_backbone.parquet`; species-specific and integrated `h5ad` objects; an `h5mu` object if any paired multiome is assembled; ranked GRN edge tables in `parquet`; candidate regulatory elements in `BED`; sequence-model scores in `parquet`; browser tracks in `bigWig`; motif and TE enrichment tables in `tsv`; and a simple interactive atlas built from `h5ad` or `zarr`. The central ranked output should be a **candidate short list** with columns for ortholog status, cell-state specificity, GRN support, chromatin support, sequence-model score, TE annotation, convergence score, and validation priority.

## Infrastructure and governance

For compute, a realistic **Phase 1** estimate is modest if you stay matrix-first: roughly **0.5–1.5 TB** of storage for processed objects and intermediate artifacts. If you decide to reprocess a large fraction of raw FASTQs, plan more conservatively for **5–10 TB**. For training and fine-tuning, a single **40 GB GPU** is enough for most scGPT or scvi-tools jobs on stromal subsets; two 40–80 GB GPUs are more comfortable if you keep broad multi-species cell sets in memory or run repeated sequence-model rescoring. CPU needs are moderate for preprocessing but can rise sharply for repeated motif scans and large permutation tests, so plan for at least **32–64 cores** and **256–512 GB RAM** on the main compute node. These are planning estimates, based on the public dataset sizes and the scale of the models being used. <sup>51</sup>

For reproducibility, use a workflow manager such as **Snakemake** or **Nextflow**, package the environment in **Docker**, lock Python and R environments, and version datasets and intermediate tables with **DVC** or an equivalent artifact tracker. Keep one metadata manifest that records accession, version date, genome build, cell filtering choices, ortholog table version, and model checkpoint hash. In this project, reproducibility failures are more likely to come from silent annotation drift and accession updates than from the models themselves.

On ethical and legal constraints, treat human reproductive datasets conservatively. Some human uterine resources are fully open, while others may route through controlled-access archives such as EGA; the human endometrium mapping paper explicitly notes that some in vivo snRNA-seq data were deposited to EGA. Also check software and model licenses before standardizing the stack: AlphaGenome is available in preview via an API for non-commercial research, which may not be suitable for all deployment contexts. More generally, do not mix public human reproductive data with any re-identifiable metadata, and keep redistribution limited to the permissions of the source repositories and software licenses. <sup>52</sup>

## Open questions and limitations

The main limitation is biological, not computational: in the sources reviewed, **public bat and spiny-mouse datasets do not yet mirror the richness of the human and mouse endometrial resources**, especially for cycle-staged chromatin and single-cell uterine profiling. That means the AI phase can produce a **high-quality shortlist of convergent regulators and elements**, but it cannot by itself close the case on convergence in the strict evolutionary sense without either broader outgroup inclusion or new focal-species data generation. <sup>53</sup>

The second limitation is model-domain mismatch. Geneformer is strongest in human single-cell settings; LINGER is strongest with paired multiome; and AlphaGenome, while highly promising, introduces access and reproducibility considerations. The safest way to manage that risk is to make all major conclusions depend on **agreement across at least one cell-state model, one GRN model, and one open sequence model** before a candidate advances to validation. <sup>54</sup>

The final implementation recommendation is therefore straightforward: **start with human + mouse + bat, use spiny mouse as a sequence-first species in Phase 1, and make the primary output a ranked, evidence-weighted candidate list rather than a claim of solved convergence**. That will give a skilled developer and computational biologist a tractable phase with a clear stopping point, and it will leave the project in the right shape for a follow-on phase that either adds new spiny-mouse uterine data or expands the phylogenetic test set. <sup>55</sup>

---

<sup>1</sup> <https://pmc.ncbi.nlm.nih.gov/articles/PMC3528014/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3528014/>

<sup>2</sup> <sup>29</sup> <sup>30</sup> <sup>45</sup> <https://www.nature.com/articles/s41592-024-02201-0>

<https://www.nature.com/articles/s41592-024-02201-0>

<sup>3</sup> <sup>25</sup> <https://www.mdpi.com/2073-4425/11/8/935>

<https://www.mdpi.com/2073-4425/11/8/935>

<sup>4</sup> <sup>48</sup> <https://academic.oup.com/mbe/article/38/3/1060/5981116>

<https://academic.oup.com/mbe/article/38/3/1060/5981116>

<sup>5</sup> <sup>9</sup> <sup>51</sup> <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE183771>

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE183771>

<sup>6</sup> <sup>55</sup> An integrated single-cell reference atlas of the human ...

[https://www.nature.com/articles/s41588-024-01873-w?utm\\_source=chatgpt.com](https://www.nature.com/articles/s41588-024-01873-w?utm_source=chatgpt.com)

<sup>7</sup> GSE111976 - GEO Accession viewer - NIH

[https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE111976&utm\\_source=chatgpt.com](https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE111976&utm_source=chatgpt.com)

<sup>8</sup> <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE127918>

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE127918>

<sup>10</sup> <sup>50</sup> <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE205477>

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE205477>

<sup>11</sup> <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE205473>

<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE205473>

- 12 <https://www.omicsdi.org/dataset/geo/GSE62475>  
<https://www.omicsdi.org/dataset/geo/GSE62475>
- 13 <https://pmc.ncbi.nlm.nih.gov/articles/PMC8648563/>  
<https://pmc.ncbi.nlm.nih.gov/articles/PMC8648563/>
- 14 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE180485>  
<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE180485>
- 15 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE168405>  
<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE168405>
- 16 [https://0-www-ncbi-nlm-nih-gov.brum.beds.ac.uk/assembly/GCF\\_021234435.1](https://0-www-ncbi-nlm-nih-gov.brum.beds.ac.uk/assembly/GCF_021234435.1)  
[https://0-www-ncbi-nlm-nih-gov.brum.beds.ac.uk/assembly/GCF\\_021234435.1](https://0-www-ncbi-nlm-nih-gov.brum.beds.ac.uk/assembly/GCF_021234435.1)
- 17 <https://github.com/CoyneLabDuke/Jamaican-Fruit-Bat-Placental-Tissue-and-Organoids>  
<https://github.com/CoyneLabDuke/Jamaican-Fruit-Bat-Placental-Tissue-and-Organoids>
- 18 28 53 <https://www.sciencedirect.com/science/article/pii/S2211124725014172>  
<https://www.sciencedirect.com/science/article/pii/S2211124725014172>
- 19 21 [https://www.ncbi.nlm.nih.gov/datasets/genome/GCA\\_029890205.1/](https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_029890205.1/)  
[https://www.ncbi.nlm.nih.gov/datasets/genome/GCA\\_029890205.1/](https://www.ncbi.nlm.nih.gov/datasets/genome/GCA_029890205.1/)
- 20 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM1054986>  
<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSM1054986>
- 22 38 <https://www.omicsdi.org/dataset/biostudies-arrayexpress/E-MTAB-11734>  
<https://www.omicsdi.org/dataset/biostudies-arrayexpress/E-MTAB-11734>
- 23 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE226417>  
<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE226417>
- 24 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE226429>  
<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE226429>
- 26 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE205480>  
<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE205480>
- 27 <https://www.omicsdi.org/dataset/biostudies-arrayexpress/E-MTAB-12105>  
<https://www.omicsdi.org/dataset/biostudies-arrayexpress/E-MTAB-12105>
- 31 54 <https://pubmed.ncbi.nlm.nih.gov/37258680/>  
<https://pubmed.ncbi.nlm.nih.gov/37258680/>
- 32 <https://docs.scvi-tools.org/>  
<https://docs.scvi-tools.org/>
- 33 39 <https://www.nature.com/articles/s41587-024-02182-7>  
<https://www.nature.com/articles/s41587-024-02182-7>
- 34 46 49 <https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0012776>  
<https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0012776>
- 35 47 <https://www.nature.com/articles/s41592-021-01252-x>  
<https://www.nature.com/articles/s41592-021-01252-x>

36 <https://www.nature.com/articles/s41588-024-02053-6>

<https://www.nature.com/articles/s41588-024-02053-6>

37 <https://www.nature.com/articles/s41586-025-10014-0>

<https://www.nature.com/articles/s41586-025-10014-0>

40 <https://www.ensembl.org/Help/View?id=135>

<https://www.ensembl.org/Help/View?id=135>

41 <https://www.ensembl.org/info/data/biomart/index.html>

<https://www.ensembl.org/info/data/biomart/index.html>

42 <https://cdr.lib.unc.edu/downloads/j098zj10z>

<https://cdr.lib.unc.edu/downloads/j098zj10z>

43 <https://www.10xgenomics.com/support/software/cell-ranger/8.0/algorithms-overview/cr-gex-algorithm>

<https://www.10xgenomics.com/support/software/cell-ranger/8.0/algorithms-overview/cr-gex-algorithm>

44 <https://scanpy.readthedocs.io/>

<https://scanpy.readthedocs.io/>

52 <https://www.nature.com/articles/s41588-021-00972-2>

<https://www.nature.com/articles/s41588-021-00972-2>